Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

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***Abstract*—We study online control of a scalar storage system with nonnegative adversarial resource arrivals and state-dependent action constraints. The adversarial, time-varying cost depends on both state and action. At each time, the controller selects a feasible action before the current resource arrival and cost are revealed. To handle the coupling between feasibility and online learning, we introduce *Simplex Disturbance-Action Control* (SDAC), whose policies have H nonnegative parameters summing to at most one. Every fixed SDAC policy is feasible by construction. Our online SDAC controller updates these parameters using entropic online mirror descent while preserving feasibility along the time-varying trajectory. We prove a regret bound of $O\Big(\sqrt{T\log(H+1)}\Big)$ relative to the best fixed SDAC policy, and prove a lower bound showing that the dependence on T is tight for fixed H. We also introduce infinite-memory SDAC policies. SDAC policies approximate this class with an error that decreases geometrically with H. For a suitable choice $H=\Theta(\log T)$, the resulting regret against infinite-memory SDAC policies is $O\big(\sqrt{T\log\log T}\big)$. The framework applies to energy-harvesting batteries and other storage-constrained resource systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state-dependent action constraints.**

I. Introduction

A. Motivation and Problem Setting

MANY resource-allocation systems have a storage state that changes over time and limits the control action at each time. An action that yields an immediate benefit or reward may leave less resource for future decisions. On the other hand, an overly conservative resource allocation policy can miss many opportunities for reward. The controller must balance these effects while satisfying the action constraint at every time.

We study this problem for a scalar storage system that loses a fixed fraction of its stored resource each period. For example, an imperfect battery can lose some of its charge even if it is not used. At each time t, the controller observes the current resource state x_t and chooses an amount u_t to use. This incurs a cost that depends on both x_t and u_t. Rewards can be modeled as negative costs. The cost $c_t(x_t,u_t)$ is revealed *after* the decision u_t is made. The cost sequence and the sequence of new resource arrivals may be chosen adversarially, so we do not assume a known probability model. Our goal is to construct an online controller whose cumulative cost is close to that of the best SDAC policy, chosen in hindsight, from a specified benchmark class.

One difficulty is that the current action affects the next state. In online control, *Disturbance-Action Control* (DAC) parameterizes actions using a linear filter of past disturbances, enabling efficient algorithms with bounded regret. In our storage model, however, some DAC policies can produce negative actions or actions larger than the available resource.

We address this issue by introducing *Simplex Disturbance-Action Control* (SDAC). SDAC restricts the filter parameters to a subprobability simplex and includes powers of the retention coefficient in the filter. These choices ensure that every fixed SDAC policy satisfies the action constraint. The state and action along each SDAC-policy trajectory are also affine in the policy parameter. Thus, a convex state–action cost is convex in that parameter, allowing updates through online convex optimization.

B. Model and Information Pattern

The storage state $x_t \geq 0$ evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t,$$

where $\alpha \in (0,1)$ is a fixed retention coefficient, u_t is the control action, and $w_t \in [0,1]$ is a nonnegative resource arrival. The action must satisfy

$$0 \leq u_t \leq \alpha x_t.$$

Thus, the controller cannot use more resource than remains after storage loss.

At time t, the controller chooses u_t using the current state and the information available through time $t-1$. The arrival w_t and the convex cost function c_t are revealed after the action is selected, and the controller incurs the cost $c_t(x_t,u_t)$. Both the online trajectory and every SDAC-policy trajectory used as a benchmark must satisfy the same dynamics and action constraint. The formal model and assumptions are given in Section II.

Our main application is energy-harvesting battery management, where x_t is the battery state of charge, w_t is the harvested energy, u_t is the energy used, and $1-\alpha$ is the fraction of stored energy lost per period. Similar dynamics arise in perishable inventory systems, water reservoirs with storage loss, and queues with abandonment. In many standard queueing models, $\alpha = 1$; that case is outside our main results and is discussed in Section VII.

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C. Contributions

Our contributions are:

- **Policy class.** We introduce SDAC policies parameterized by a subprobability simplex. We prove that every such SDAC policy is feasible for every arrival sequence with entries in $[0, 1]$.
  We also introduce infinite-memory SDAC policies in Section III-C. Their parameter domain extends the subprobability simplex to nonnegative sequences whose sum is at most one. This class contains every SDAC policy and every feasible fixed-fraction policy $u_t = kx_t$ with $k \in [0, \alpha]$. Truncating the parameter sequence gives an SDAC policy whose approximation error decreases geometrically with the memory horizon H.
- **Online control and regret.** We develop an online SDAC controller that updates its parameter using entropic online mirror descent (OMD) over the subprobability simplex. We prove that the controller remains feasible and achieves $O\Big(\sqrt{T\log(H+1)}\Big)$ regret against the best fixed SDAC policy. For a suitable choice $H = \Theta(\log T)$, the regret against infinite-memory SDAC policies is $O\big(\sqrt{T\log\log T}\big)$. The same bound holds against the best fixed-fraction policy.
- **Lower bound.** We prove a lower bound of order $\sqrt{T}$ for the expected regret of every causal feasible controller, including randomized controllers; see Theorem 2. When T is at least on the order of $(1-\alpha)^{-1}$, the lower bound scales as $\Omega((1-\alpha)^{-1}\sqrt{T})$ near $\alpha = 1$. The upper bound contains a factor of $\sqrt{\log(H+1)}$ that does not appear in the lower bound. The bounds also differ by a factor of $(1-\alpha)^{-1/2}$ in their dependence on $1-\alpha$. Subsequent work closes these gaps for the same policy class [1].

D. Related Work

a) Online control: Goulart et al. use a parameterization similar to DAC for a related constrained control problem under bounded disturbances [2]. Agarwal et al. use DAC policies to obtain $\widetilde{O}(\sqrt{T})$ regret against strongly stable linear controllers [3]. Golowich et al. use simplex constraints on DAC parameters and entropic mirror descent to control population dynamics [4]. Their feasible action set is independent of the current state. In our storage model, the feasible action interval $[0, \alpha x_t]$ changes with the state. SDAC combines a subprobability simplex with the factors α^i accounting for storage loss to ensure that every fixed SDAC policy satisfies this constraint for every admissible arrival sequence. When the policy parameters change online, our controller clips the action to preserve feasibility. Our regret analysis bounds the additional cost caused by parameter updates and clipping.

Li et al. study online control with fixed state and action constraints [5]. Their method requires a linear controller that stays strictly within the constraints. This condition can fail here: when $x_t = 0$, feasibility requires $u_t = 0$. Liu et al. study online nonstochastic control with static and adversarial constraints [6]. They allow joint state–action constraints, including our fixed affine constraint $0 \le u_t \le \alpha x_t$. Their $\widetilde{O}(\sqrt{T})$ regret guarantee requires a fixed DAC policy that satisfies the constraints with a uniform positive margin. Their projection method for avoiding constraint violations also relies on strict feasibility.

This margin is unavailable in our model. If $x_1 = 0$ and $w_t = 0$ for every t, then $x_t = u_t = 0$ at every time, so both inequalities in the action constraint hold with equality.

Without strict feasibility, their result allows constraint violations. Our controller satisfies the action constraint at every time and achieves $O(\sqrt{T\log(H+1)})$ regret against the best fixed SDAC policy, without assuming strict feasibility.

Online convex optimization with memory studies costs that depend on past decisions [7], [8]. We use a related approach: analyze costs with the SDAC parameter held fixed, then bound the additional cost caused by online updates. Our analysis accounts for storage loss and action clipping (Lemma 5). Kumar et al. also give lower bounds when the effect of past decisions decays geometrically [8]. Our lower bound applies to causal feasible controllers, with fixed SDAC policies as the benchmark (Section VI).

b) Energy harvesting: Fixed-fraction policies for stochastic arrivals motivate our infinite-memory SDAC class. Shaviv and Özgür [9] study these policies under i.i.d. energy arrivals. Arafa et al. [10] extend the analysis to increasing concave rewards. Zibaeenejad and Chen [11] study allocation with knowledge of upcoming arrivals under a Bernoulli model in which each nonzero arrival fully recharges the battery. For our storage dynamics, infinite-memory SDAC includes every feasible fixed-fraction policy (Corollary 1).

Yu and Neely [12] and Asgari and Neely [13] study online energy allocation under stochastic arrivals. In the latter work, each arrival is known before allocation and the cost is revealed afterward. Both works compare expected performance to fixed actions whose energy use does not exceed the mean arrival. These benchmark actions need not be feasible on every arrival sequence, and the costs do not depend directly on the battery state. Our fixed SDAC policies remain feasible on their own trajectories, with actions that depend on the arrival history.

Lin et al. [14] study online energy allocation with arbitrary arrivals and a class of time-varying concave rewards. They give competitive-ratio guarantees against an offline optimum and observe the current arrival and reward before acting. In our setting, the arrival and cost are revealed after the action, and convex costs may depend on both the storage state and the action.

c) Storage management: Qin et al. study storage control with energy loss and capacity constraints under stochastic assumptions [15]. They observe the current price and net demand before acting and give guarantees for long-run average cost. Kim et al. study a related storage problem and give competitive guarantees for the value of meeting demand [16]. Both works account for limited storage capacity. Our analysis assumes enough capacity to avoid overflow and studies regret under adversarial arrivals and costs revealed after the action.

d) Queueing and network optimization: Classical queueing control studies throughput and queue stability [17]–[19]. Stochastic network optimization studies tradeoffs between average performance and average queue backlog [20].

Khojastepour and Sabharwal [21] use filters similar to SDAC for scheduling with a delay constraint. Their fixed filters combine current and past arrivals using nonnegative coefficients summing to one. They study these filters under i.i.d. arrivals and also develop a time-varying policy that does not require the arrival distribution. SDAC uses parameters summing to at most one, accounts for storage loss, and acts before the current arrival is observed. We study regret under adversarial convex costs depending on both state and action.

e) Inventory and reservoir management: Yang et al. [22] study online inventory control with linear costs and capacity and rate constraints. They obtain optimal competitive-ratio guarantees against an offline optimum. They observe the current demand and price before acting, whereas our controller acts before the current arrival and cost are revealed.

Hihat et al. [23] allow non-i.i.d. demands and prove feasibility and $O(\sqrt{T})$ expected regret against the best fixed inventory target. Their analysis requires probabilistic assumptions on demand. Our arrivals are arbitrary, and our benchmark is a fixed SDAC policy. Hihat and Fermanian [24] study inventory systems with multiple products, capacity constraints, and delivery delays. They evaluate their method empirically and leave regret guarantees open. Classical reservoir theory studies storage under random arrivals [25].

II. Model and Regret Benchmark

For every positive integer n, we write $[n] \triangleq \{1, \dots, n\}$. For every $z \in \mathbb{R}$, we define $[z]_+ \triangleq \max\{z, 0\}$. For $\mathbf{v} \in \mathbb{R}^n$, the i-th coordinate is denoted by v_i. If the vector already carries a subscript, the coordinate is appended after a comma: the i-th coordinate of $\mathbf{v}_t$ is $v_{t,i}$. We write $\mathbf{1}$ for the all-ones vector, with dimension clear from context, and $\langle \cdot, \cdot \rangle$ for the Euclidean inner product. We use $\log$ for the natural logarithm.

Fix a positive integer T and a known retention coefficient $\alpha \in (0, 1)$. The state $x_t \in \mathbb{R}_{\geq 0}$ is the resource level at the beginning of time t. The system starts empty:

$$x_1 = 0. \tag{1}$$

At each time $t \in [T]$, the controller chooses an action $u_t \in \mathbb{R}_{\geq 0}$ satisfying

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives an arrival $w_t \in [0, 1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

Here, $1 - \alpha$ is the fraction of stored resource lost per period. The constraint (2) ensures that the action does not exceed the amount remaining after this loss.

In the online-control literature, w_t is commonly called a disturbance. Here, we call it an *arrival* because it is nonnegative, while retaining the standard name Disturbance-Action Control. Nonnegativity is important for both the storage interpretation and the feasibility properties of SDAC.

At time t, the controller chooses u_t before observing w_t or c_t. After the action is selected, the arrival and the entire cost function are revealed, the controller incurs $c_t(x_t, u_t)$, and the state is updated according to (3).

For every feasible trajectory, (2), (3), and $w_t \in [0, 1]$ give

$$0 \leq x_{t+1} \leq \alpha x_t + 1. \tag{4}$$

Induction using (1) and (4), with $0 < \alpha < 1$, gives

$$\begin{aligned} 0 \leq x_t &\leq \sum_{s=0}^{t-2} \alpha^s = \frac{1 - \alpha^{t-1}}{1 - \alpha} \\ &\leq \frac{1}{1 - \alpha}, \qquad t \in [T + 1]. \end{aligned} \tag{5}$$

Equations (2) and (5) place every feasible state–action pair in the compact set

$$\mathcal{D}_\alpha \triangleq \left\{ (x, u) : 0 \leq x \leq \frac{1}{1 - \alpha}, 0 \leq u \leq \alpha x \right\}. \tag{6}$$

Remark 1 (Storage capacity). A storage capacity of $1/(1-\alpha)$ suffices to avoid overflow.

Each revealed cost function $c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$ is convex in (x, u) and satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there are constants $L_x, L_u > 0$ such that

$$|c_t(x, u) - c_t(x', u')| \leq L_x |x - x'| + L_u |u - u'| \tag{7}$$

for all $(x, u), (x', u') \in \mathcal{D}_\alpha$. At every $(x, u) \in \mathcal{D}_\alpha$, we also assume access to a cost subgradient

$$\boldsymbol{\zeta}_t(x, u) = \big(\zeta_t^x(x, u), \zeta_t^u(x, u)\big) \in \partial c_t(x, u) \tag{8}$$

satisfying

$$|\zeta_t^x(x, u)| \leq L_x, \qquad |\zeta_t^u(x, u)| \leq L_u. \tag{9}$$

These subgradient bounds hold, for example, when c_t has a convex extension to an open neighborhood of $\mathcal{D}_\alpha$ satisfying (7); see [26, Thm. 9.13].

We use x_t and u_t without parameter arguments for the trajectory generated by the online controller. Its cumulative cost is

$$J_T^{\mathrm{on}} \triangleq \sum_{t=1}^{T} c_t(x_t, u_t). \tag{10}$$

The superscript on stands for online.

Section III defines a class of SDAC policies characterized by a parameter $\mathbf{m}$ in a parameter set $\mathcal{M}_H$. For a fixed $\mathbf{m} \in \mathcal{M}_H$, the corresponding *SDAC-policy trajectory* is the state–action sequence $\{(x_t(\mathbf{m}), u_t(\mathbf{m}))\}_{t=1}^T$ obtained by applying the SDAC policy with parameter $\mathbf{m}$ at every time, starting from $x_1(\mathbf{m}) = 0$. Its cumulative cost is

$$J_T(\mathbf{m}) \triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big). \tag{11}$$

The online policy in (10) and the fixed SDAC policy in (11) are evaluated on the same arrival sequence $\{w_t\}_{t=1}^T$ and cost function sequence $\{c_t\}_{t=1}^T$. For a fixed positive integer H, the regret against the best fixed SDAC policy is

$$\mathrm{Reg}_T^{\mathrm{SDAC}}(H) \triangleq J_T^{\mathrm{on}} - \min_{\mathbf{m} \in \mathcal{M}_H} J_T(\mathbf{m}). \tag{12}$$

Our objective is to satisfy (2) at every time and achieve sublinear regret.

III. Simplex Disturbance-Action Control (SDAC)

A. From Disturbance-Action Control to SDAC

DAC uses a finite linear filter of past disturbances. Under standard conditions, this parameterization yields convex surrogate costs and efficient online updates. In our scalar storage model, DAC has the following form:

$$u_t = \sum_{i=1}^{H} \theta_i w_{t-i}, \tag{13}$$

where H is a positive integer (the memory horizon) and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_H)^\top \in \mathbb{R}^H$ is the vector of filter coefficients.

For our constrained model, an unrestricted filter does not ensure feasibility: its action can exceed αx_t or become negative. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which uses a simplex-constrained filter tailored to the storage dynamics. This restriction makes every fixed SDAC policy feasible while keeping surrogate costs convex and online updates efficient.

B. Policies and Structural Properties

Fix a memory horizon $H \in \mathbb{N}_{>0}$ and define the subprobability simplex

$$\mathcal{M}_H \triangleq \left\{ \mathbf{m} \in \mathbb{R}^H : m_i \geq 0, \quad \sum_{i=1}^{H} m_i \leq 1 \right\}. \tag{14}$$

The inequalities $m_i \geq 0$ hold for every $i \in [H]$. We write $\|\mathbf{m}\|_1 = \sum_{i=1}^{H} |m_i|$ for the ℓ_1 norm. With the convention $w_\tau \equiv 0$ for $\tau \leq 0$, each $\mathbf{m} \in \mathcal{M}_H$ defines an SDAC policy through

$$u_t(\mathbf{m}) \triangleq \sum_{i=1}^{H} m_i \alpha^i w_{t-i}. \tag{15}$$

Here, $m_i \alpha^i w_s$ is the amount from arrival w_s used i periods later.

The following state decomposition yields two structural properties: every fixed SDAC policy satisfies the feasibility constraint (2), and its induced cost is convex and Lipschitz in $\mathbf{m}$. Omitted proofs appear in Appendix A.

Lemma 1 (State decomposition under SDAC)**.** *Fix a positive integer H. For $t \in [T]$ and $i \in [H]$, define*

$$\begin{aligned} r_t(i) &\triangleq \sum_{j=1}^{i} \alpha^{j-1} w_{t-j}, \\ r_t &\triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j}, \end{aligned} \tag{16}$$

where the empty sum defining r_1 is zero. Then, for every $\mathbf{m} \in \mathcal{M}_H$ and $t \in [T]$, the SDAC-policy state associated with $\mathbf{m}$ satisfies

$$x_t(\mathbf{m}) = \sum_{i=1}^{H} m_i r_t(i) + \big(1 - \sum_{i=1}^{H} m_i\big) r_t. \tag{17}$$

By (17) and (15), the state and action of an SDAC policy can be written together as

$$\begin{aligned} x_t(\mathbf{m}) &= r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle, \\ u_t(\mathbf{m}) &= \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle, \end{aligned} \tag{18}$$

where the coefficient vectors in $\mathbb{R}^H$ are

$$\begin{aligned} \boldsymbol{\phi}_t^x &\triangleq \big(r_t(1) - r_t, \dots, r_t(H) - r_t\big)^\top, \\ \boldsymbol{\phi}_t^u &\triangleq \big(\alpha w_{t-1}, \alpha^2 w_{t-2}, \dots, \alpha^H w_{t-H}\big)^\top. \end{aligned} \tag{19}$$

The superscripts x and u identify the coefficients for the state and action, respectively. Both vectors and r_t depend only on past arrivals and α, so they are independent of $\mathbf{m}$. Thus, both the state and action are affine in $\mathbf{m}$.

Lemma 2 (Feasibility of SDAC policies)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the fixed SDAC policy indexed by $\mathbf{m}$ is feasible:*

$$0 \leq u_t(\mathbf{m}) \leq \alpha x_t(\mathbf{m}) \qquad \text{for every } t \in [T]. \tag{20}$$

Proof. Fix $t \in [T]$. By (16) and $w_\tau \geq 0$,

$$\alpha r_t(i) - \alpha^i w_{t-i} = \sum_{j=1}^{i-1} \alpha^j w_{t-j} \geq 0, \qquad i \in [H]. \tag{21}$$

Using (14), $r_t \geq 0$ from (16), and (15),

$$\begin{aligned} 0 \leq u_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \\ &\overset{(21)}{\leq} \sum_{i=1}^{H} m_i \alpha r_t(i) \\ &\leq \sum_{i=1}^{H} m_i \alpha r_t(i) + \alpha \left(1 - \sum_{i=1}^{H} m_i\right) r_t \\ &\overset{(17)}{=} \alpha x_t(\mathbf{m}). \end{aligned}$$

This proves (20). □

Lemma 3 (Convexity and Lipschitz continuity of the surrogate cost)**.** *For $t \in [T]$, define*

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H, \tag{22}$$

and

$$G_\alpha \triangleq \alpha \left(\frac{L_x}{1-\alpha} + L_u \right). \tag{23}$$

Then ℓ_t is convex on $\mathcal{M}_H$ and, for all $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| \leq G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1. \tag{24}$$

For $\mathbf{m} \in \mathcal{M}_H$, use (8) *to define*

$$\begin{aligned} (\zeta_t^x, \zeta_t^u) &= \boldsymbol{\zeta}_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \\ \mathbf{g}_t &\triangleq \zeta_t^x \boldsymbol{\phi}_t^x + \zeta_t^u \boldsymbol{\phi}_t^u. \end{aligned} \tag{25}$$

Then

$$\mathbf{g}_t \in \partial \ell_t(\mathbf{m}), \tag{26}$$

$$\|\mathbf{g}_t\|_\infty \leq G_\alpha. \tag{27}$$

C. Approximation of Infinite-Memory SDAC Policies

Define the infinite-memory parameter domain

$$\mathcal{M}_\infty \triangleq \left\{ \mathbf{m} = (m_i)_{i\ge 1} : m_i \ge 0, \quad \sum_{i=1}^{\infty} m_i \le 1 \right\}. \tag{28}$$

Each $\mathbf{m} \in \mathcal{M}_\infty$ defines an infinite-memory SDAC policy through

$$u_t(\mathbf{m}) \triangleq \sum_{i=1}^{t-1} m_i \alpha^i w_{t-i}. \tag{29}$$

The state $x_t(\mathbf{m})$ follows (3) with initial condition (1). We use (11) to define $J_T(\mathbf{m})$ for $\mathbf{m} \in \mathcal{M}_\infty$ as well. After extending vectors in $\mathcal{M}_H$ by zeros, SDAC policies with horizon H are exactly the infinite-memory SDAC policies with $m_i = 0$ for every $i > H$.

Lemma 4 (Approximation of infinite-memory SDAC policies)**.** *Every infinite-memory SDAC policy is feasible for every arrival sequence with entries in* $[0,1]$*. Fix* $\mathbf{m} \in \mathcal{M}_\infty$ *and an integer* $H \ge 1$*, and define its truncation by*

$$\mathbf{m}_H(\mathbf{m}) \triangleq (m_1, \dots, m_H)^\top. \tag{30}$$

Then $\mathbf{m}_H(\mathbf{m}) \in \mathcal{M}_H$*. If each* c_t *satisfies* (7)*, then, for every such arrival sequence,*

$$\begin{aligned} &\left| J_T\big(\mathbf{m}_H(\mathbf{m})\big) - J_T(\mathbf{m}) \right| \\ &\qquad \le \left(L_u + \frac{L_x}{1-\alpha} \right) \alpha^{H+1} T. \end{aligned} \tag{31}$$

A *fixed-fraction policy*, parameterized by $k \in [0, \alpha]$, chooses $u_t = k x_t$ at every time. We denote its trajectory by $\{(x_t(k), u_t(k))\}_{t=1}^T$, where

$$\begin{aligned} u_t(k) &= k x_t(k), \qquad x_1(k) = 0, \\ x_{t+1}(k) &= (\alpha - k) x_t(k) + w_t. \end{aligned} \tag{32}$$

Since $\alpha - k \ge 0$ and $w_t \ge 0$, induction gives $x_t(k) \ge 0$. Thus, $0 \le u_t(k) = k x_t(k) \le \alpha x_t(k)$, so every such policy is feasible.

Corollary 1 (Fixed-fraction policies as infinite-memory SDAC policies)**.** *For* $k \in [0, \alpha]$*, define*

$$m_i(k) = \frac{k}{\alpha} \left(1 - \frac{k}{\alpha} \right)^{i-1}, \qquad i \ge 1, \tag{33}$$

where $0^0 = 1$*. Then* $\mathbf{m}(k) \in \mathcal{M}_\infty$ *and, for every arrival sequence with entries in* $[0,1]$*, the infinite-memory SDAC policy indexed by* $\mathbf{m}(k)$ *coincides with the feasible fixed-fraction policy* $u_t(k) = k x_t(k)$*:*

$$\big(x_t(\mathbf{m}(k)), u_t(\mathbf{m}(k))\big) = \big(x_t(k), u_t(k)\big), \qquad t \in [T]. \tag{34}$$

IV. Online SDAC Controller

The controller maintains a parameter $\mathbf{m}_t \in \mathcal{M}_H$ and updates it by OMD [27] with step size $\eta > 0$ and unnormalized negentropy:

$$\Psi(\mathbf{m}) \triangleq \sum_{i=1}^{H} \big(m_i \log m_i - m_i \big), \qquad \mathbf{m} \in \mathbb{R}^H_{\ge 0}, \tag{35}$$

where $0 \log 0 \triangleq 0$. For $\mathbf{m} \in \mathbb{R}^H_{\ge 0}$ and $\mathbf{v} \in \mathbb{R}^H_{>0}$, the associated Bregman divergence is

$$D_\Psi(\mathbf{m} \| \mathbf{v}) \triangleq \sum_{i=1}^{H} \Big(m_i \log \tfrac{m_i}{v_i} - m_i + v_i \Big). \tag{36}$$

Initialize the online parameter as

$$\mathbf{m}_1 \triangleq \frac{1}{H+1} \mathbf{1}. \tag{37}$$

a) Action selection: At time t, the controller uses $\mathbf{m}_t$, initialized by (37) for $t = 1$ and produced by the OMD update at time $t-1$ for $t \ge 2$. The feasibility guarantee for an SDAC policy uses the same parameter at every time. When $\mathbf{m}_t$ varies, the nominal action $\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle$ can exceed αx_t. The controller therefore selects

$$u_t \triangleq \min\{ \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle, \alpha x_t \}. \tag{38}$$

The state then evolves according to (3). Since $\boldsymbol{\phi}_t^u$ and $\mathbf{m}_t$ are componentwise nonnegative, $\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle \ge 0$. If $x_t \ge 0$, then (38) gives $0 \le u_t \le \alpha x_t$, and (3) gives $x_{t+1} \ge w_t \ge 0$. Since $x_1 = 0$, induction shows that the online trajectory is feasible.

Equation (94) shows that the D_Ψ-projection of $\mathbf{m}_t$ onto the parameters compatible with x_t induces exactly the clipped action u_t.

b) Surrogate cost and OMD update: After w_t and c_t are revealed, the controller evaluates the subgradient $\mathbf{g}_t$ in (25) at $\mathbf{m}_t$ and sets

$$\mathbf{m}_{t+1} \triangleq \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} \{ \eta \langle \mathbf{g}_t, \mathbf{m} \rangle + D_\Psi(\mathbf{m} \| \mathbf{m}_t) \}. \tag{39}$$

Lemma 8 gives a two-step implementation. First, take the coordinatewise multiplicative step

$$\widetilde{m}_{t+1,i} \triangleq m_{t,i} \exp\big(-\eta g_{t,i}\big), \qquad i \in [H]. \tag{40}$$

Next, project onto $\mathcal{M}_H$ by rescaling if needed:

$$\begin{aligned} \mathbf{m}_{t+1} &= \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} D_\Psi(\mathbf{m} \| \widetilde{\mathbf{m}}_{t+1}) \\ &= \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}. \end{aligned} \tag{41}$$

V. Regret Upper Bounds

The next theorem bounds the regret of the online SDAC controller against the best fixed SDAC policy. Proofs appear in Appendix C.

Theorem 1 (Regret against fixed SDAC policies)**.** *Suppose the costs satisfy the assumptions of Section II. For any step size* $\eta > 0$*, the controller in Section IV, initialized by* (37)*, satisfies*

$$\mathrm{Reg}_T^{\mathrm{SDAC}}(H) \le \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha, \tag{42}$$

where

$$\Gamma_\alpha \triangleq \frac{G_\alpha^2}{2} + \frac{\alpha G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}, \tag{43}$$

and $G_\alpha = \alpha\big(\frac{L_x}{1-\alpha} + L_u\big)$ *as in* (23)*.*

Choosing

$$\eta^* = \sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}} \tag{44}$$

gives

$$\mathrm{Reg}_T^{\mathrm{SDAC}}(H) \le 2\sqrt{\Gamma_\alpha T \log(H+1)}. \tag{45}$$

The proof combines the OMD bound for the surrogate costs (Lemma 10) with the following bound on the difference between realized and surrogate costs.

Lemma 5 (Stability under online parameter updates)**.** *Under the assumptions of Theorem 1, the online SDAC controller satisfies*

$$\sum_{t=1}^{T}\Big(c_t(x_t,u_t) - \ell_t(\mathbf{m}_t)\Big) \le \frac{\alpha\eta T G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}. \tag{46}$$

Corollary 2 (No-regret against infinite-memory SDAC policies)**.** *Under the assumptions of Theorem 1, run the online SDAC controller with step size η^* and memory horizon*

$$H = H_T \triangleq \max\left\{1, \left\lceil \frac{\log T}{\log(1/\alpha)} \right\rceil \right\}, \tag{47}$$

where η^* is given by (44) *with $H = H_T$. Define the regret against the best fixed infinite-memory SDAC policy by*

$$\mathrm{Reg}_T^{\mathrm{SDAC}}(\infty) \triangleq J_T^{\mathrm{on}} - \inf_{\mathbf{m}\in\mathcal{M}_\infty} J_T(\mathbf{m}). \tag{48}$$

Then

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{SDAC}}(\infty) \le{}& 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &+ \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T.\end{aligned} \tag{49}$$

For fixed α, L_x, L_u, the right-hand side is $O\big(\sqrt{T\log\log T}\big) = o(T)$. The same bound holds against the best fixed-fraction policy, since Corollary 1 gives

$$J_T^{\mathrm{on}} - \min_{k\in[0,\alpha]} \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big) \le \mathrm{Reg}_T^{\mathrm{SDAC}}(\infty). \tag{50}$$

VI. Minimax Regret Lower Bound

We now show that, with H, α, L_x, L_u fixed, no causal feasible controller can guarantee worst-case expected regret growing more slowly than $\sqrt{T}$. Thus, the upper bound in Theorem 1 has the best possible dependence on T, up to a constant factor. The lower bound applies to every causal feasible controller; only the hindsight benchmark is restricted to SDAC policies.

Theorem 2 (SDAC regret lower bound)**.** *Fix $\alpha \in (0,1)$, $L_x, L_u > 0$, and an integer $H \ge 1$. Assume the dynamics, initial state, and feedback of Section II, and let the integer horizon satisfy*

$$T \ge 2\left\lceil \frac{\log 2}{\log(1/\alpha)} \right\rceil + 2. \tag{51}$$

For every causal feasible online controller $\mathcal{A}$, possibly randomized, there is a deterministic sequence of arrivals and costs, fixed before the interaction, that satisfies the model assumptions with constants L_x, L_u and gives

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}(H)\big] \ge \frac{G_\alpha}{8}\sqrt{T}, \tag{52}$$

where $G_\alpha = \alpha(L_x/(1-\alpha) + L_u)$ as in (23)*. The expectation is over the controller's internal randomization. The sequence may depend on $\mathcal{A}$, but not on its realized random choices.*

The upper bound (45) is $2\sqrt{\Gamma_\alpha T\log(H+1)}$. For fixed H, α, L_x, L_u, both bounds therefore scale as $\sqrt{T}$. The upper bound contains a factor $\sqrt{\log(H+1)}$, whereas the lower bound is independent of H. Appendix D proves the theorem and also gives a lower bound valid for every $T \ge 2$ (Lemma 11).

Remark 2 (Large-α regime). For fixed $L_x, L_u > 0$, as $\alpha \uparrow 1$, both G_α and the right-hand side of (51) are of order $(1-\alpha)^{-1}$. Under this horizon condition, the lower bound is $\Omega((1-\alpha)^{-1}\sqrt{T})$, while the upper bound is $O((1-\alpha)^{-3/2}\sqrt{T\log(H+1)})$. Thus, these bounds differ by factors of $\sqrt{\log(H+1)}$ in memory dependence and $(1-\alpha)^{-1/2}$ in retention dependence. Subsequent work closes these gaps for the same policy class [1].

This comparison applies under the stated horizon condition. For fixed T, the right-hand side of the general bound (137) remains finite as $\alpha \uparrow 1$. The calculations are given in Appendix D.

Lower-bound construction: Set $w_t = 1$ and compare the fixed SDAC parameters $\mathbf{m}^{(0)} = \mathbf{0}$ and $\mathbf{m}^{(1)} = \mathbf{e}_1 = (1,0,\dots,0)$. The first policy never uses resource; the second uses each arrival one period later, after storage loss. For $t = 2,\dots,T$, their trajectories are

$$\begin{aligned} u_t(\mathbf{m}^{(0)}) &= 0, & u_t(\mathbf{m}^{(1)}) &= \alpha, \\ x_t(\mathbf{m}^{(0)}) &= \frac{1-\alpha^{t-1}}{1-\alpha}, & x_t(\mathbf{m}^{(1)}) &= 1. \end{aligned}$$

Both parameters belong to $\mathcal{M}_H$ for every $H \ge 1$, which explains why the lower bound does not depend on H.

Choose costs $c_t(x,u) = \xi_t(L_x x - L_u u)$, where the signs ξ_t are independent, equally likely to be 1 or -1, and independent of the controller's randomization. Causality gives every online controller zero expected cost. Each fixed comparison policy also has zero expected cost, but choosing the better of the two in hindsight gives an advantage. Their cumulative cost difference is the weighted sum $\sum_{t=2}^{T}\xi_t\beta_t$ in (143), with β_t defined in (138). Khintchine's inequality [28] bounds the expected absolute value of this sum from below. Averaging over the signs then gives a deterministic sequence with the stated expected regret.

VII. Discussion and Extensions

We discuss three extensions. In each case, the feasibility, convexity, and stability arguments apply with the stated changes to the policy domain and Lipschitz constants. We omit the resulting regret bounds, whose form is unchanged up to these constants and, in the vector settings, the chosen product norm.

a) Retention $\alpha \geq 1$: Here we extend the model beyond the assumption $\alpha \in (0,1)$. The case $\alpha = 1$ has no storage loss, while $\alpha > 1$ represents growth between decisions. Fix k with $\alpha - 1 < k < \alpha$, so $k > 0$, and define

$$\begin{aligned} \bar{\alpha} &\triangleq \alpha - k \in (0,1), \\ v_t &\triangleq u_t - kx_t, \qquad 0 \leq v_t \leq \bar{\alpha} x_t. \end{aligned} \tag{53}$$

Substituting (53) into (3) gives

$$\begin{aligned} x_{t+1} &= \alpha x_t - (kx_t + v_t) + w_t \\ &= (\alpha - k)x_t - v_t + w_t = \bar{\alpha} x_t - v_t + w_t. \end{aligned} \tag{54}$$

Thus, v_t is the control action for a storage system with retention coefficient $\bar{\alpha} \in (0,1)$. For $\mathbf{m} \in \mathcal{M}_H$, define

$$\begin{aligned} v_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \bar{\alpha}^i w_{t-i}, \\ u_t(\mathbf{m}) &= kx_t(\mathbf{m}) + v_t(\mathbf{m}), \end{aligned} \tag{55}$$

where $x_t(\mathbf{m})$ follows (54) with initial condition (1). Equation (20), with α replaced by $\bar{\alpha}$, gives

$$0 \leq v_t(\mathbf{m}) \leq \bar{\alpha} x_t(\mathbf{m}). \tag{56}$$

For $\mathbf{m}_t \in \mathcal{M}_H$, the online controller instead selects

$$\begin{aligned} v_t &= \min\left\{ \sum_{i=1}^{H} m_{t,i} \bar{\alpha}^i w_{t-i},\ \bar{\alpha} x_t \right\}, \\ u_t &= kx_t + v_t. \end{aligned} \tag{57}$$

Equations (56) and (57) give $0 \leq v_t \leq \bar{\alpha} x_t$ for the fixed and online policies, respectively. Using $k > 0$ and (53),

$$\begin{aligned} 0 \leq kx_t \leq u_t &= kx_t + v_t \\ &\leq (k + \bar{\alpha}) x_t \\ &= \alpha x_t, \end{aligned}$$

so the original feasibility constraint (2) is satisfied. Equation (5), with α replaced by $\bar{\alpha} \in (0,1)$, applies to (54) and gives $0 \leq x_t \leq 1/(1-\bar{\alpha})$. Assume that the original costs $c_t(x,u)$ are convex and satisfy the conditions (7), (8), and (9) on

$$0 \leq x \leq \frac{1}{1-\bar{\alpha}}, \qquad kx \leq u \leq \alpha x. \tag{58}$$

The cost in the variables (x,v) is $c_t(x, kx+v)$, which is convex because $(x,v) \mapsto (x, kx+v)$ is affine. For two feasible pairs (x,v) and (x',v'), (7) on (58) and the triangle inequality give

$$\begin{aligned} &|c_t(x, kx+v) - c_t(x', kx'+v')| \\ &\quad \leq L_x |x - x'| + L_u |k(x-x') + v - v'| \\ &\quad \leq L_x |x - x'| + kL_u |x - x'| + L_u |v - v'| \\ &\quad = (L_x + kL_u)|x - x'| + L_u |v - v'|. \end{aligned} \tag{59}$$

Let

$$(\zeta_t^x, \zeta_t^u) = \boldsymbol{\zeta}_t(x, kx+v). \tag{60}$$

By (8) and (60),

$$\begin{aligned} &c_t(x', kx'+v') - c_t(x, kx+v) \\ &\quad \geq \zeta_t^x (x' - x) + \zeta_t^u \big(k(x'-x) + v' - v\big) \\ &\quad = (\zeta_t^x + k\zeta_t^u)(x' - x) + \zeta_t^u (v' - v). \end{aligned} \tag{61}$$

Thus, (61) shows that $(\zeta_t^x + k\zeta_t^u, \zeta_t^u)$ is a subgradient of the transformed cost. The triangle inequality and (9) give

$$\begin{aligned} |\zeta_t^x + k\zeta_t^u| \leq |\zeta_t^x| + k|\zeta_t^u| \leq L_x + kL_u, \\ |\zeta_t^u| \leq L_u. \end{aligned} \tag{62}$$

The online controller uses the same OMD update with surrogate cost

$$\mathbf{m} \mapsto c_t\big(x_t(\mathbf{m}), kx_t(\mathbf{m}) + v_t(\mathbf{m})\big). \tag{63}$$

By (59) and (62), the feasibility and regret upper-bound analysis applies to (63) with α replaced by $\bar{\alpha}$ and L_x replaced by $L_x + kL_u$, while L_u is unchanged. Both the online controller and the comparators use the same fixed k. Each comparator uses (55). The substitution (53) preserves the cost at every time, so it also preserves cumulative costs and regret against this class. Selecting k, either in advance or online, is left for future work.

b) Vector actions and multi-task allocation: Consider a single storage state x_t with a vector action $\mathbf{u}_t \in \mathbb{R}^n_{\geq 0}$ that allocates resources across n tasks, subject to $\sum_{a=1}^n u_{t,a} \leq \alpha x_t$. The dynamics and initial state are

$$x_{t+1} = \alpha x_t - \sum_{a=1}^{n} u_{t,a} + w_t, \qquad x_1 = 0, \tag{64}$$

and the cost $c_t(x, \mathbf{u})$ is convex in $(x, \mathbf{u})$. Replace $\mathbf{m} \in \mathcal{M}_H$ with a matrix $\mathbf{M} \in \mathbb{R}^{n \times H}_{\geq 0}$ satisfying

$$\begin{aligned} \sum_{a=1}^{n} \sum_{i=1}^{H} M_{a,i} &\leq 1, \\ u_{t,a}(\mathbf{M}) &= \sum_{i=1}^{H} M_{a,i} \alpha^i w_{t-i}. \end{aligned} \tag{65}$$

By nonnegativity and (65),

$$\begin{aligned} m_i &\triangleq \sum_{a=1}^{n} M_{a,i} \geq 0, \qquad i \in [H], \\ \sum_{i=1}^{H} m_i &= \sum_{a=1}^{n} \sum_{i=1}^{H} M_{a,i} \leq 1. \end{aligned} \tag{66}$$

Thus, $\mathbf{m} \in \mathcal{M}_H$ by (14). Summing the actions in (65) gives

$$\begin{aligned} \sum_{a=1}^{n} u_{t,a}(\mathbf{M}) &= \sum_{i=1}^{H} \left(\sum_{a=1}^{n} M_{a,i} \right) \alpha^i w_{t-i} \\ &\overset{(66)}{=} \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \\ &\overset{(15)}{=} u_t(\mathbf{m}). \end{aligned} \tag{67}$$

By (67), (64), (3), and (1), the two state trajectories have the same total action and initial state. Hence,

$$x_t(\mathbf{M}) = x_t(\mathbf{m}). \tag{68}$$

Therefore,

$$\begin{aligned} \sum_{a=1}^{n} u_{t,a}(\mathbf{M}) &\overset{(67)}{=} u_t(\mathbf{m}) \\ &\overset{(20)}{\leq} \alpha x_t(\mathbf{m}) \\ &\overset{(68)}{=} \alpha x_t(\mathbf{M}). \end{aligned}$$

By (18), (66), and (68), the state is affine in $\mathbf{M}$. The vector action is affine by (65), so $c_t\big(x_t(\mathbf{M}), \mathbf{u}_t(\mathbf{M})\big)$ is convex in $\mathbf{M}$. For the online SDAC controller, a nominal vector whose entries sum to more than αx_t can be scaled proportionally so that its entries remain nonnegative and sum to αx_t. Under an ℓ_1 Lipschitz bound in the action, the size of this correction equals the excess total action, and the stability proof proceeds with the corresponding matrix norm. OMD then runs over the matrix subprobability simplex in (65).

c) Multiple storage states with diagonal dynamics: Suppose the state is $\mathbf{x}_t \in \mathbb{R}^d_{\geq 0}$ and evolves according to $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t - \mathbf{u}_t + \mathbf{w}_t$, where $\mathbf{A} = \operatorname{diag}(\alpha_1, \dots, \alpha_d)$, $\alpha_s \in (0,1)$ for $s \in [d]$, the arrivals satisfy $\mathbf{w}_t \in [0,1]^d$, and the per-storage constraints are $0 \leq u_{t,s} \leq \alpha_s x_{t,s}$. Since $\mathbf{A}$ is diagonal,

$$x_{t+1,s} = \alpha_s x_{t,s} - u_{t,s} + w_{t,s}, \qquad s \in [d]. \tag{69}$$

In (69), the next state of each storage system depends only on its own current state, action, and arrival. Assign one SDAC parameter in $\mathcal{M}_H$ to each storage state and use (15) with α_s^i in coordinate s. The dynamics and feasibility constraints decouple by storage state, and the SDAC parameter domain is a product of subprobability simplices. With (1) in each coordinate, (5), with α replaced by α_s, gives $x_{t,s} \leq 1/(1-\alpha_s)$ for every $s \in [d]$. The online SDAC controller clips each nominal action to $\alpha_s x_{t,s}$. If the cost is jointly convex and satisfies compatible Lipschitz and bounded-subgradient assumptions on the resulting bounded product domain, the surrogate cost is jointly convex in all SDAC parameters and OMD applies on the product set. With a sum of negentropy regularizers, the OMD minimization separates by storage state, although each subgradient may depend on all storage states and actions. Cross-coupled storage systems (nondiagonal $\mathbf{A}$) require further analysis.

VIII. Conclusion

We studied adversarial online control of a storage system under a state-dependent action constraint. We introduced SDAC policies, which are feasible by construction, and developed an online SDAC controller with $O\Big(\sqrt{T\log(H+1)}\Big)$ regret against the best fixed SDAC policy. The SDAC approximation results further give $O\big(\sqrt{T\log\log T}\big)$ regret against the best fixed infinite-memory SDAC policy for a suitable choice $H = \Theta(\log T)$. Under the horizon condition (51), the lower bound gives $\Omega((1-\alpha)^{-1}\sqrt{T})$ regret as $\alpha \uparrow 1$. For the bounds proved here, a factor of $\sqrt{\log(H+1)}$ remains between the upper and lower bounds in their dependence on H, together with a factor of $(1-\alpha)^{-1/2}$ in their dependence on $1-\alpha$. Subsequent work closes these gaps for the same policy class [1]. The model applies directly to energy-harvesting battery management and related settings in which storage limits must be respected in real time.

Disclosure of AI Assistance

The lower-bound construction and proofs in Section VI and Appendix D were developed primarily using ChatGPT (5.6 sol). The construction closely follows the argument based on independent random signs in [29, Sec. 2.3, footnote 3], adapted to the storage dynamics and feasible SDAC policies. The remaining material was originally written by the authors. ChatGPT (5.6 sol) was subsequently used to improve its exposition and to refine constants in the proofs.

Appendix A
Proofs for Simplex Disturbance-Action Control

Proof of Lemma 1. By (1), (16), and $w_\tau = 0$ for $\tau \leq 0$,

$$x_1(\mathbf{m}) = r_1(i) = r_1 = 0, \qquad i \in [H]. \tag{70}$$

For $1 \leq t < T$ and $i \in [H]$, shifting the summation index in (16) gives

$$\begin{aligned} r_{t+1} &= \sum_{j=1}^{t} \alpha^{j-1} w_{t+1-j} = w_t + \sum_{j=1}^{t-1} \alpha^j w_{t-j} \\ &= w_t + \alpha r_t, \\ r_{t+1}(i) &= w_t + \sum_{j=1}^{i-1} \alpha^j w_{t-j} \\ &= w_t + \alpha r_t(i) - \alpha^i w_{t-i}. \end{aligned} \tag{71}$$

The sums in (71) are empty when their upper limit is zero.

Suppose (17) holds at $1 \leq t < T$. Substituting it and (15) into (3) gives

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \sum_{i=1}^{H} m_i\big(\alpha r_t(i) - \alpha^i w_{t-i}\big) \\ &\quad + \left(1 - \sum_{i=1}^{H} m_i\right) \alpha r_t + w_t \\ &= \sum_{i=1}^{H} m_i\big(\alpha r_t(i) - \alpha^i w_{t-i} + w_t\big) \\ &\quad + \left(1 - \sum_{i=1}^{H} m_i\right) (\alpha r_t + w_t) \\ &\overset{(71)}{=} \sum_{i=1}^{H} m_i r_{t+1}(i) \\ &\quad + \left(1 - \sum_{i=1}^{H} m_i\right) r_{t+1}. \end{aligned}$$

Together with (70), this proves (17) by induction. □

Proof of Lemma 3. Fix $t \in [T]$. By (18) and (22), ℓ_t is the composition of the convex function c_t with an affine map, so it is convex. Equations (20) and (5) place every SDAC-policy state–action pair in $\mathcal{D}_\alpha$.

By (16) and (19), using $0 \le w_\tau \le 1$ and $0 < \alpha < 1$,

$$\begin{aligned}\|\boldsymbol{\phi}_t^x\|_\infty &= \max_{i\in[H]}\big(r_t - r_t(i)\big)\\ &= \max_{i\in[H]} \sum_{j=i+1}^{t-1} \alpha^{j-1} w_{t-j}\\ &\le \max_{i\in[H]} \sum_{j=i+1}^{\infty} \alpha^{j-1}\\ &= \max_{i\in[H]} \frac{\alpha^i}{1-\alpha} = \frac{\alpha}{1-\alpha},\\ \|\boldsymbol{\phi}_t^u\|_\infty &= \max_{i\in[H]} \alpha^i w_{t-i}\\ &\le \max_{i\in[H]} \alpha^i = \alpha.\end{aligned} \tag{72}$$

The finite sum in (72) is empty when $i \ge t-1$, since $w_\tau = 0$ for $\tau \le 0$.

Fix $\mathbf{m} \in \mathcal{M}_H$, and let (ζ_t^x, ζ_t^u) and $\mathbf{g}_t$ be defined by (25). For every $\mathbf{m}' \in \mathcal{M}_H$, (8) gives

$$\begin{aligned}&\ell_t(\mathbf{m}') - \ell_t(\mathbf{m})\\ &\quad\ge \zeta_t^x\big(x_t(\mathbf{m}') - x_t(\mathbf{m})\big)\\ &\qquad + \zeta_t^u\big(u_t(\mathbf{m}') - u_t(\mathbf{m})\big)\\ &\quad\overset{(18)}{=} \big\langle \zeta_t^x \boldsymbol{\phi}_t^x + \zeta_t^u \boldsymbol{\phi}_t^u, \mathbf{m}' - \mathbf{m}\big\rangle\\ &\quad\overset{(25)}{=} \langle \mathbf{g}_t, \mathbf{m}' - \mathbf{m}\rangle.\end{aligned} \tag{73}$$

Thus, (73) proves (26).

By the triangle inequality,

$$\begin{aligned}\|\mathbf{g}_t\|_\infty &\le |\zeta_t^x| \|\boldsymbol{\phi}_t^x\|_\infty + |\zeta_t^u| \|\boldsymbol{\phi}_t^u\|_\infty\\ &\overset{(9)}{\le} L_x \|\boldsymbol{\phi}_t^x\|_\infty + L_u \|\boldsymbol{\phi}_t^u\|_\infty\\ &\overset{(72)}{\le} \frac{\alpha L_x}{1-\alpha} + \alpha L_u\\ &\overset{(23)}{=} G_\alpha.\end{aligned}$$

This proves (27).

By (73), Hölder's inequality, and (27),

$$\begin{aligned}\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}') &\le \langle \mathbf{g}_t, \mathbf{m} - \mathbf{m}'\rangle\\ &\le \|\mathbf{g}_t\|_\infty \|\mathbf{m} - \mathbf{m}'\|_1\\ &\le G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.\end{aligned} \tag{74}$$

Applying (74) in both orders of $\mathbf{m}, \mathbf{m}'$ proves (24). □

Proof of Lemma 4. By (30) and (28),

$$\begin{aligned}&m_i \ge 0, \qquad i \in [H],\\ &\|\mathbf{m}_H(\mathbf{m})\|_1 = \sum_{i=1}^{H} m_i \le \sum_{i=1}^{\infty} m_i \le 1.\end{aligned} \tag{75}$$

Hence, $\mathbf{m}_H(\mathbf{m}) \in \mathcal{M}_H$ by (14).

Take $H = T$ in (75). Since $w_\tau = 0$ for $\tau \le 0$, (15) and (29) give, for $t \in [T]$,

$$\begin{aligned}u_t\big(\mathbf{m}_T(\mathbf{m})\big) &= \sum_{i=1}^{t-1} m_i \alpha^i w_{t-i}\\ &= u_t(\mathbf{m}),\\ x_t\big(\mathbf{m}_T(\mathbf{m})\big) &= x_t(\mathbf{m}).\end{aligned} \tag{76}$$

The state equality in (76) follows by induction using (1) and (3). By (76) and (20), the infinite-memory policy is feasible.

For any $H \ge 1$, define

$$\begin{aligned}e_t^u &\triangleq u_t(\mathbf{m}) - u_t\big(\mathbf{m}_H(\mathbf{m})\big),\\ e_t^x &\triangleq x_t\big(\mathbf{m}_H(\mathbf{m})\big) - x_t(\mathbf{m}).\end{aligned} \tag{77}$$

By (15) and (29), using $0 \le w_\tau \le 1$ and $\alpha^i \le \alpha^{H+1}$ for $i > H$,

$$\begin{aligned}0 \le e_t^u &= \sum_{i=H+1}^{t-1} \alpha^i m_i w_{t-i}\\ &\le \sum_{i=H+1}^{t-1} \alpha^i m_i\\ &\le \alpha^{H+1} \sum_{i=H+1}^{t-1} m_i\\ &\overset{(28)}{\le} \alpha^{H+1}.\end{aligned} \tag{78}$$

The sums in (78) are empty when $t \le H+1$.

Subtracting (3) for the two policies and using (1) and (77) gives

$$e_1^x = 0, \qquad e_{t+1}^x = \alpha e_t^x + e_t^u. \tag{79}$$

Iterating (79), with $0 < \alpha < 1$,

$$\begin{aligned}0 \le e_t^x &= \sum_{s=1}^{t-1} \alpha^{t-1-s} e_s^u\\ &\overset{(78)}{\le} \alpha^{H+1} \sum_{s=1}^{t-1} \alpha^{t-1-s}\\ &= \alpha^{H+1} \sum_{j=0}^{t-2} \alpha^j\\ &\le \alpha^{H+1} \sum_{j=0}^{\infty} \alpha^j = \frac{\alpha^{H+1}}{1-\alpha}.\end{aligned} \tag{80}$$

The finite sums in (80) are empty at $t = 1$.

Feasibility and (5) place both state–action pairs in $\mathcal{D}_\alpha$. By (7) and the nonnegative errors in (78) and (80),

$$\begin{aligned}&\Big|c_t\big(x_t(\mathbf{m}_H(\mathbf{m})), u_t(\mathbf{m}_H(\mathbf{m}))\big)\\ &\qquad - c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big)\Big|\\ &\quad\le L_x e_t^x + L_u e_t^u\\ &\quad\le \frac{L_x \alpha^{H+1}}{1-\alpha} + L_u \alpha^{H+1}.\end{aligned} \tag{81}$$

Summing (81) and applying the triangle inequality to (11) proves (31). □

Proof of Corollary 1. Since $0 \le k/\alpha \le 1$, (33) gives $m_i(k) \ge 0$ for every $i \ge 1$ and

$$\begin{aligned}\sum_{i=1}^{H} m_i(k) &= \sum_{i=1}^{H} \left[\left(1 - \frac{k}{\alpha}\right)^{i-1} - \left(1 - \frac{k}{\alpha}\right)^{i}\right]\\ &= 1 - \left(1 - \frac{k}{\alpha}\right)^{H} \le 1,\\ \sum_{i=1}^{\infty} m_i(k) &= \lim_{H\to\infty} \sum_{i=1}^{H} m_i(k) \le 1.\end{aligned} \tag{82}$$

Thus, (82) and (28) give $\mathbf{m}(k) \in \mathcal{M}_\infty$.

Iterating the fixed-fraction recursion (32) gives

$$x_t(k) = \sum_{i=1}^{t-1} (\alpha - k)^{i-1} w_{t-i}. \tag{83}$$

For every $i \geq 1$, (33) and $\alpha > 0$ give

$$\begin{aligned} \alpha^i m_i(k) &= \alpha^i \frac{k}{\alpha} \left(\frac{\alpha - k}{\alpha} \right)^{i-1} \\ &= k(\alpha - k)^{i-1}. \end{aligned} \tag{84}$$

By (32), (83), and (84),

$$\begin{aligned} u_t(k) &= k x_t(k) \\ &= \sum_{i=1}^{t-1} k(\alpha - k)^{i-1} w_{t-i} \\ &= \sum_{i=1}^{t-1} \alpha^i m_i(k) w_{t-i} \\ &\overset{(29)}{=} u_t\big(\mathbf{m}(k)\big). \end{aligned} \tag{85}$$

Induction using (1), (3), and (85) proves (34). □

APPENDIX B
PROOFS FOR THE ONLINE SDAC CONTROLLER

The OMD results for unnormalized negentropy and its Bregman divergence on the probability simplex are standard [27]. We prove the corresponding forms for the subprobability simplex $\mathcal{M}_H$.

Lemma 6 (Strong convexity of negentropy on the subprobability simplex)**.** *For every $\mathbf{m} \in \mathcal{M}_H$ and $\mathbf{v} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$,*

$$D_\Psi(\mathbf{m} \| \mathbf{v}) \geq \frac{1}{2} \|\mathbf{m} - \mathbf{v}\|_1^2. \tag{86}$$

Thus, Ψ is 1-strongly convex with respect to $\|\cdot\|_1$ on $\mathcal{M}_H$.

Proof. First suppose $\mathbf{m} > 0$ coordinatewise. For $s \in [0,1]$, define

$$\begin{aligned} \mathbf{d} &\triangleq \mathbf{m} - \mathbf{v}, \\ \mathbf{z}_s &\triangleq \mathbf{v} + s\mathbf{d} \\ &= (1-s)\mathbf{v} + s\mathbf{m} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}. \end{aligned} \tag{87}$$

By (35),

$$\nabla^2 \Psi(\mathbf{z}_s) = \operatorname{diag}(1/z_{s,1}, \ldots, 1/z_{s,H}). \tag{88}$$

Cauchy–Schwarz and (14) give

$$\begin{aligned} \|\mathbf{d}\|_1^2 &= \left(\sum_{i=1}^H \frac{|d_i|}{\sqrt{z_{s,i}}} \sqrt{z_{s,i}} \right)^2 \\ &\leq \left(\sum_{i=1}^H \frac{d_i^2}{z_{s,i}} \right) \left(\sum_{i=1}^H z_{s,i} \right) \\ &\overset{(14)}{\leq} \sum_{i=1}^H \frac{d_i^2}{z_{s,i}} \\ &\overset{(88)}{=} \mathbf{d}^\top \nabla^2 \Psi(\mathbf{z}_s) \mathbf{d}. \end{aligned} \tag{89}$$

By Taylor's formula along (87) and (36),

$$\begin{aligned} D_\Psi(\mathbf{m} \| \mathbf{v}) &= \int_0^1 (1-s) \mathbf{d}^\top \nabla^2 \Psi(\mathbf{z}_s) \mathbf{d} \, ds \\ &\overset{(89)}{\geq} \|\mathbf{d}\|_1^2 \int_0^1 (1-s) \, ds \\ &\overset{(87)}{=} \frac{1}{2} \|\mathbf{m} - \mathbf{v}\|_1^2. \end{aligned} \tag{90}$$

For any $\mathbf{m} \in \mathcal{M}_H$ and $0 < \varepsilon < 1$, define

$$\begin{aligned} \mathbf{m}^\varepsilon &\triangleq (1-\varepsilon)\mathbf{m} + \varepsilon \mathbf{v} \\ &\in \mathcal{M}_H \cap \mathbb{R}^H_{>0}. \end{aligned} \tag{91}$$

By (36) and $0 \log 0 = 0$, the divergence is continuous in its first argument. Therefore,

$$\begin{aligned} D_\Psi(\mathbf{m} \| \mathbf{v}) &= \lim_{\varepsilon \downarrow 0} D_\Psi(\mathbf{m}^\varepsilon \| \mathbf{v}) \\ &\overset{(90)}{\geq} \lim_{\varepsilon \downarrow 0} \frac{1}{2} \|\mathbf{m}^\varepsilon - \mathbf{v}\|_1^2 \\ &\overset{(91)}{=} \lim_{\varepsilon \downarrow 0} \frac{1}{2} (1-\varepsilon)^2 \|\mathbf{m} - \mathbf{v}\|_1^2 \\ &= \frac{1}{2} \|\mathbf{m} - \mathbf{v}\|_1^2. \end{aligned}$$ □

Lemma 7 (Action clipping as a Bregman projection)**.** *For $t \in [T]$, define the set of parameters compatible with the current state by*

$$\mathcal{B}_t \triangleq \{\mathbf{m} \in \mathcal{M}_H : \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle \leq \alpha x_t\}. \tag{92}$$

Let $\mathbf{m}_t^{\text{exec}}$ be the Bregman projection of $\mathbf{m}_t$ onto $\mathcal{B}_t$:

$$\mathbf{m}_t^{\text{exec}} \in \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{B}_t} D_\Psi(\mathbf{m} \| \mathbf{m}_t). \tag{93}$$

Then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}} \rangle = \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle, \alpha x_t\} = u_t. \tag{94}$$

Proof. By (37), (40), and (41), every coordinate of $\mathbf{m}_t$ is positive. By (1), (3), and (38), induction gives $x_t \geq 0$. Thus, $\mathbf{0} \in \mathcal{B}_t$ by (92). The set $\mathcal{B}_t$ is compact and convex, and $D_\Psi(\cdot \| \mathbf{m}_t)$ is continuous and convex by (36). Hence, (93) has a minimizer. By (86) and (36), for every $\mathbf{m} \in \mathcal{M}_H$,

$$\begin{aligned} D_\Psi(\mathbf{m} \| \mathbf{m}_t) &\geq \frac{1}{2} \|\mathbf{m} - \mathbf{m}_t\|_1^2 \geq 0, \\ D_\Psi(\mathbf{m}_t \| \mathbf{m}_t) &= 0. \end{aligned} \tag{95}$$

If $\mathbf{m}_t \in \mathcal{B}_t$, then (93) and (95) give

$$\mathbf{m}_t^{\text{exec}} = \mathbf{m}_t. \tag{96}$$

Otherwise, suppose for contradiction that

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}} \rangle < \alpha x_t < \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle. \tag{97}$$

By (97),

$$\delta \triangleq \frac{\alpha x_t - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}} \rangle}{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}} \rangle} \in (0,1). \tag{98}$$

Convexity of $\mathcal{M}_H$ gives

$$\mathbf{m}_\delta \triangleq (1-\delta)\mathbf{m}_t^{\text{exec}} + \delta \mathbf{m}_t \in \mathcal{M}_H. \tag{99}$$

Using (99) and (98),

$$\begin{aligned}&\langle \boldsymbol{\phi}_t^u, \mathbf{m}_\delta\rangle\\ &= \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle\\ &\quad + \delta\left(\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle\right)\\ &= \alpha x_t.\end{aligned} \tag{100}$$

Thus, (92) and (100) give $\mathbf{m}_\delta \in \mathcal{B}_t$. By (97) and (95),

$$D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t) \geq \frac{1}{2}\|\mathbf{m}_t^{\mathrm{exec}} - \mathbf{m}_t\|_1^2 > 0. \tag{101}$$

Convexity in the first argument and (99) give

$$\begin{aligned}D_\Psi(\mathbf{m}_\delta\|\mathbf{m}_t) &\leq (1-\delta)D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t)\\ &\quad + \delta D_\Psi(\mathbf{m}_t\|\mathbf{m}_t)\\ &\overset{(95)}{=} (1-\delta)D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t)\\ &\overset{(98)}{\underset{(101)}{<}} D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t),\end{aligned}$$

contrary to (93). Hence, when $\mathbf{m}_t \notin \mathcal{B}_t$,

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \alpha x_t. \tag{102}$$

Combining (96) and (102) with (38) proves (94). □

Lemma 8 (Closed-form solution of the OMD update)**.** *Fix $\mathbf{m}_t \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$, $\eta > 0$, and $\mathbf{g}_t \in \mathbb{R}^H$. Define $\widetilde{\mathbf{m}}_{t+1}$ by* (40)*. The unique solution of* (39) *is given by* (41) *and satisfies*

$$\mathbf{m}_{t+1} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}. \tag{103}$$

Thus, the iterates initialized by (37) *remain strictly positive.*

Proof. By (40), $\widetilde{m}_{t+1,i} > 0$ for every $i \in [H]$. For $\mathbf{m} \in \mathcal{M}_H$, (36) and (40) give

$$\begin{aligned}&\eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\\ &= \sum_{i=1}^H \left(m_i \log\frac{m_i}{\widetilde{m}_{t+1,i}} - m_i + m_{t,i}\right)\\ &= D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) + \sum_{i=1}^H (m_{t,i} - \widetilde{m}_{t+1,i}).\end{aligned} \tag{104}$$

Terms with $m_i = 0$ use $0\log 0 = 0$ throughout. The term $\sum_{i=1}^H (m_{t,i} - \widetilde{m}_{t+1,i})$ in (104) is independent of $\mathbf{m}$.

Define

$$\mathbf{m}^\star \triangleq \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}. \tag{105}$$

Since $\widetilde{\mathbf{m}}_{t+1} > 0$ coordinatewise,

$$\begin{aligned}m_i^\star > 0, \qquad i \in [H],\\ \|\mathbf{m}^\star\|_1 = \frac{\|\widetilde{\mathbf{m}}_{t+1}\|_1}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}} \leq 1.\end{aligned} \tag{106}$$

For $i \in [H]$, (105) gives

$$\log\frac{m_i^\star}{\widetilde{m}_{t+1,i}} = -\log\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}. \tag{107}$$

If $\|\widetilde{\mathbf{m}}_{t+1}\|_1 \leq 1$, then $\log\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\} = 0$; otherwise, (106) gives $\|\mathbf{m}^\star\|_1 = 1$. Thus,

$$\left(1 - \|\mathbf{m}^\star\|_1\right)\log\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\} = 0. \tag{108}$$

For $\mathbf{m} \in \mathcal{M}_H$, expand (36):

$$\begin{aligned}&D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) - D_\Psi(\mathbf{m}^\star\|\widetilde{\mathbf{m}}_{t+1})\\ &= D_\Psi(\mathbf{m}\|\mathbf{m}^\star)\\ &\quad + \sum_{i=1}^H (m_i - m_i^\star)\log\frac{m_i^\star}{\widetilde{m}_{t+1,i}}\\ &\overset{(107)}{=} D_\Psi(\mathbf{m}\|\mathbf{m}^\star)\\ &\quad + \left(\|\mathbf{m}^\star\|_1 - \|\mathbf{m}\|_1\right)\log\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}\\ &\overset{(108)}{=} D_\Psi(\mathbf{m}\|\mathbf{m}^\star)\\ &\quad + \left(1 - \|\mathbf{m}\|_1\right)\log\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}.\end{aligned} \tag{109}$$

In (109), $1 - \|\mathbf{m}\|_1 \geq 0$ by (14), and $\log\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\} \geq 0$. Using (106),

$$\begin{aligned}&D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) - D_\Psi(\mathbf{m}^\star\|\widetilde{\mathbf{m}}_{t+1})\\ &\overset{(109)}{\geq} D_\Psi(\mathbf{m}\|\mathbf{m}^\star)\\ &\overset{(86)}{\geq} \frac{1}{2}\|\mathbf{m} - \mathbf{m}^\star\|_1^2.\end{aligned} \tag{110}$$

Equations (104) and (110) give the unique minimizer $\mathbf{m}_{t+1} = \mathbf{m}^\star$ in (41). Equation (106) proves (103); induction from (37) gives positivity of every iterate. □

APPENDIX C
PROOFS FOR THE REGRET UPPER BOUNDS

Lemma 9 (One-step OMD movement bound)**.** *The OMD update* (39) *with step size $\eta > 0$ satisfies*

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \leq \eta\|\mathbf{g}_t\|_\infty \leq \eta G_\alpha. \tag{111}$$

Proof. By (37) and (103), both iterates are positive. First-order optimality in (39), using (36), gives, for every $\mathbf{m} \in \mathcal{M}_H$,

$$\begin{aligned}&\eta\langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle\\ &\quad \leq \sum_{i=1}^H (m_i - m_{t+1,i})\log\frac{m_{t+1,i}}{m_{t,i}}.\end{aligned} \tag{112}$$

Apply (86) in both directions, then (112) with $\mathbf{m} = \mathbf{m}_t$:

$$\begin{aligned}&\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2\\ &\quad \leq D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) + D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1})\\ &\quad \overset{(36)}{=} \sum_{i=1}^H (m_{t+1,i} - m_{t,i})\log\frac{m_{t+1,i}}{m_{t,i}}\\ &\quad \overset{(112)}{\leq} \eta\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle\\ &\quad \overset{\text{Hölder}}{\leq} \eta\|\mathbf{g}_t\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.\end{aligned} \tag{113}$$

If $\mathbf{m}_{t+1} = \mathbf{m}_t$, (111) holds with zero left-hand side. Otherwise, divide (113) by $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1$ and apply (27). □

Lemma 10 (OMD regret for the surrogate costs)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the update* (39) *satisfies*

$$\sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) \leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2. \tag{114}$$

Proof. By (26),

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\le \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}\rangle \\ &= \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle.\end{aligned} \tag{115}$$

By (112) and (36),

$$\begin{aligned}&\eta\langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle \\ &\le \sum_{i=1}^{H}(m_i - m_{t+1,i})\log\frac{m_{t+1,i}}{m_{t,i}} \\ &= D_\Psi(\mathbf{m}\|\mathbf{m}_t) - D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1}) \\ &\quad - D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t).\end{aligned} \tag{116}$$

Equation (116) also holds when $m_i = 0$, with $0\log 0 = 0$. Substituting (116) into (115) gives

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_t)}{\eta} - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})}{\eta} \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \\ &\quad - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta}.\end{aligned} \tag{117}$$

Since $\eta > 0$, Hölder's inequality and (86) give

$$\begin{aligned}&\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta} \\ &\overset{\text{Hölder}}{\le} \|\mathbf{g}_t\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \\ &\quad - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta} \\ &\overset{(86)}{\le} \|\mathbf{g}_t\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \\ &\quad - \frac{\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2}{2\eta} \\ &= \frac{\eta}{2}\|\mathbf{g}_t\|_\infty^2 \\ &\quad - \frac{\big(\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 - \eta\|\mathbf{g}_t\|_\infty\big)^2}{2\eta} \\ &\overset{\eta>0}{\le} \frac{\eta}{2}\|\mathbf{g}_t\|_\infty^2.\end{aligned} \tag{118}$$

Substituting (118) into (117) and summing over t gives

$$\begin{aligned}&\sum_{t=1}^{T}\big(\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\big) \\ &\le \frac{1}{\eta}\sum_{t=1}^{T}\big(D_\Psi(\mathbf{m}\|\mathbf{m}_t) - D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})\big) \\ &\quad + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2 \\ &= \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1})}{\eta} \\ &\quad + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2 \\ &\overset{(86)}{\le} \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2.\end{aligned}$$

This proves (114). □

Proof of Lemma 5. Set

$$d_t \triangleq |x_t - x_t(\mathbf{m}_t)|, \qquad d_1 \overset{(1)}{=} 0. \tag{119}$$

For $z, z' \in \mathbb{R}$,

$$|[z]_+ - [z']_+| \le |z - z'|, \qquad [z]_+ \le |z|. \tag{120}$$

For $1 \le t < T$, (18), (3), (38), and (20) give

$$\begin{aligned}x_{t+1} &= [\alpha x_t - u_t(\mathbf{m}_t)]_+ + w_t, \\ x_{t+1}(\mathbf{m}_t) &= [\alpha x_t(\mathbf{m}_t) - u_t(\mathbf{m}_t)]_+ + w_t.\end{aligned} \tag{121}$$

Equations (121) and (120) give

$$\begin{aligned}&|x_{t+1} - x_{t+1}(\mathbf{m}_t)| \\ &= \big|[\alpha x_t - u_t(\mathbf{m}_t)]_+ \\ &\qquad - [\alpha x_t(\mathbf{m}_t) - u_t(\mathbf{m}_t)]_+\big| \\ &\overset{(120)}{\le} |\alpha x_t - \alpha x_t(\mathbf{m}_t)| \overset{(119)}{=} \alpha d_t.\end{aligned} \tag{122}$$

The triangle inequality and (119) give

$$\begin{aligned}d_{t+1} &\le |x_{t+1} - x_{t+1}(\mathbf{m}_t)| \\ &\quad + |x_{t+1}(\mathbf{m}_t) - x_{t+1}(\mathbf{m}_{t+1})| \\ &\overset{\substack{(122)\\(18)}}{\le} \alpha d_t + |\langle \boldsymbol{\phi}_{t+1}^x, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle| \\ &\overset{\text{Hölder}}{\le} \alpha d_t + \|\boldsymbol{\phi}_{t+1}^x\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \\ &\overset{(72)}{\le} \alpha d_t + \frac{\alpha}{1-\alpha}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.\end{aligned} \tag{123}$$

By (119), $d_1 = 0$ and $d_T \ge 0$. Summing (123) gives

$$\begin{aligned}\sum_{t=1}^{T} d_t &= \sum_{t=1}^{T-1} d_{t+1} \\ &\le \alpha\sum_{t=1}^{T-1} d_t + \frac{\alpha}{1-\alpha}\sum_{t=1}^{T-1}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \\ &\overset{d_T\ge 0}{\le} \alpha\sum_{t=1}^{T} d_t + \frac{\alpha}{1-\alpha}\sum_{t=1}^{T-1}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.\end{aligned} \tag{124}$$

Rearranging (124), since $1 - \alpha > 0$,

$$\sum_{t=1}^{T} d_t \le \frac{\alpha}{(1-\alpha)^2}\sum_{t=1}^{T-1}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \tag{125}$$

For $T = 1$, (119) makes both sides of (125) zero.

For $t \in [T]$, use (18), (38), monotonicity of $[\cdot]_+$, and (120):

$$\begin{aligned}0 \le u_t(\mathbf{m}_t) - u_t &= [u_t(\mathbf{m}_t) - \alpha x_t]_+ \\ &\overset{(20)}{\le} [\alpha x_t(\mathbf{m}_t) - \alpha x_t]_+ \\ &\overset{(120)}{\le} \alpha|x_t(\mathbf{m}_t) - x_t| \\ &\overset{(119)}{=} \alpha d_t.\end{aligned} \tag{126}$$

Both state–action pairs are feasible, so (5) places them in $\mathcal{D}_\alpha$. By (22) and (7),

$$\begin{aligned}
&\sum_{t=1}^{T}\big(c_t(x_t,u_t)-\ell_t(\mathbf{m}_t)\big)\\
&\quad\le \sum_{t=1}^{T}\big(L_x d_t + L_u|u_t-u_t(\mathbf{m}_t)|\big)\\
&\quad\overset{(126)}{\le} (L_x+\alpha L_u)\sum_{t=1}^{T} d_t\\
&\quad\overset{(125)}{\le} \frac{\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}\sum_{t=1}^{T-1}\|\mathbf{m}_{t+1}-\mathbf{m}_t\|_1\\
&\quad\overset{(111)}{\le} \frac{\alpha\eta(T-1)G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}\\
&\quad\overset{T-1\le T}{\le} \frac{\alpha\eta T G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}.
\end{aligned}$$

This proves (46). □

Proof of Theorem 1. Fix $\mathbf{m}\in\mathcal{M}_H$. Since $H\ge 1$,

$$\log(H+1)=\int_1^{H+1}\frac{dz}{z}\ge\int_1^{H+1}\frac{dz}{H+1}=\frac{H}{H+1}. \tag{127}$$

By (37) and (36), with $0\log 0=0$,

$$\begin{aligned}
D_\Psi(\mathbf{m}\|\mathbf{m}_1)&=\sum_{i=1}^{H} m_i\log m_i\\
&\quad+\|\mathbf{m}\|_1\log(H+1)-\|\mathbf{m}\|_1+\frac{H}{H+1}\\
&\overset{(14)}{\le}\|\mathbf{m}\|_1\log(H+1)\\
&\quad+(1-\|\mathbf{m}\|_1)\frac{H}{H+1}-\frac{\|\mathbf{m}\|_1}{H+1}\\
&\overset{(127)}{\le}\|\mathbf{m}\|_1\log(H+1)\\
&\quad+(1-\|\mathbf{m}\|_1)\log(H+1)\\
&=\log(H+1).
\end{aligned} \tag{128}$$

The inequalities in (128) use $m_i\in[0,1]$ and $\|\mathbf{m}\|_1\in[0,1]$ from (14).

By (114), (128), and (27),

$$\begin{aligned}
&\sum_{t=1}^{T}\big(\ell_t(\mathbf{m}_t)-\ell_t(\mathbf{m})\big)\\
&\quad\le\frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta}+\frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2\\
&\quad\le\frac{\log(H+1)}{\eta}+\frac{\eta T G_\alpha^2}{2}.
\end{aligned} \tag{129}$$

Using (10), (11), and (22),

$$\begin{aligned}
J_T^{\mathrm{on}}-J_T(\mathbf{m})&=\sum_{t=1}^{T}\big(c_t(x_t,u_t)-\ell_t(\mathbf{m}_t)\big)\\
&\quad+\sum_{t=1}^{T}\big(\ell_t(\mathbf{m}_t)-\ell_t(\mathbf{m})\big)\\
&\overset{(46)}{\underset{(129)}{\le}}\frac{\log(H+1)}{\eta}+\frac{\eta T G_\alpha^2}{2}\\
&\quad+\frac{\alpha\eta T G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}\\
&\overset{(43)}{=}\frac{\log(H+1)}{\eta}+\eta T\Gamma_\alpha.
\end{aligned} \tag{130}$$

Taking the maximum over $\mathbf{m}\in\mathcal{M}_H$ in (130) and using (12) proves (42). Since $H,T\ge 1$ and $\Gamma_\alpha>0$, (44) gives $\eta^*>0$ and

$$\frac{\log(H+1)}{\eta^*}=\eta^* T\Gamma_\alpha=\sqrt{\Gamma_\alpha T\log(H+1)}. \tag{131}$$

Substituting (131) into (42) proves (45). □

Proof of Corollary 2. Fix $\mathbf{m}\in\mathcal{M}_\infty$. By (75) and (14), $\mathbf{m}_{H_T}(\mathbf{m})\in\mathcal{M}_{H_T}$. Hence,

$$\begin{aligned}
J_T^{\mathrm{on}}-J_T(\mathbf{m})&=J_T^{\mathrm{on}}-J_T\big(\mathbf{m}_{H_T}(\mathbf{m})\big)\\
&\quad+J_T\big(\mathbf{m}_{H_T}(\mathbf{m})\big)-J_T(\mathbf{m})\\
&\overset{(12)}{\le}\mathrm{Reg}_T^{\mathrm{SDAC}}(H_T)\\
&\quad+J_T\big(\mathbf{m}_{H_T}(\mathbf{m})\big)-J_T(\mathbf{m})\\
&\overset{(31)}{\le}\mathrm{Reg}_T^{\mathrm{SDAC}}(H_T)\\
&\quad+\left(L_u+\frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T.
\end{aligned} \tag{132}$$

Taking the supremum over $\mathbf{m}\in\mathcal{M}_\infty$ in (132), then using (48) and (45) with $H=H_T$, proves (49).

Since $T\ge 1$ and $0<\alpha<1$, (47) gives

$$\frac{\log T}{\log(1/\alpha)}\le H_T\le 1+\frac{\log T}{\log(1/\alpha)}. \tag{133}$$

In particular, $H_T=1$ when $T=1$. Since $0<\alpha<1$,

$$\begin{aligned}
\alpha^{H_T+1}T&\overset{(133)}{\le}\alpha^{1+\log T/\log(1/\alpha)}T\\
&=\alpha\exp\left(\frac{\log\alpha}{\log(1/\alpha)}\log T\right)T\\
&=\alpha e^{-\log T}T=\alpha.
\end{aligned} \tag{134}$$

By (133) and monotonicity of log, for fixed α as $T\to\infty$,

$$\begin{aligned}
\log(H_T+1)&\le\log\left(2+\frac{\log T}{\log(1/\alpha)}\right)\\
&=O(\log\log T).
\end{aligned} \tag{135}$$

For fixed α, L_x, L_u, (134) and (135) bound the right-hand side of (49) by $O\big(\sqrt{T\log\log T}\big)=o(T)$.

For every $k \in [0, \alpha]$, Corollary 1 gives $\mathbf{m}(k) \in \mathcal{M}_\infty$. Using (34) and (11),

$$\begin{aligned} &\inf_{\mathbf{m}\in\mathcal{M}_\infty} J_T(\mathbf{m}) \\ &\quad\le \min_{k\in[0,\alpha]} J_T(\mathbf{m}(k)) \\ &\quad= \min_{k\in[0,\alpha]} \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big). \end{aligned} \tag{136}$$

Subtracting (136) from J_T^{on} and using (48) proves (50). □

Appendix D
Proofs for the Regret Lower Bound

Lemma 11 (Lower bound for every $T \ge 2$)**.** *Fix $\alpha \in (0,1)$, $L_x, L_u > 0$, and integers $H \ge 1$ and $T \ge 2$. For every causal feasible online controller $\mathcal{A}$, possibly randomized, there is a deterministic oblivious sequence of arrivals and costs satisfying the assumptions of Section II such that*

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}(H)\big] \ge \frac{1}{2\sqrt{2}} \sqrt{\sum_{t=2}^{T} \beta_t^2}, \tag{137}$$

where

$$\beta_t \triangleq \alpha \left(L_u + \frac{L_x(1-\alpha^{t-2})}{1-\alpha} \right), \qquad t = 2, \dots, T. \tag{138}$$

The expectation is over the internal randomization of $\mathcal{A}$. The sequence may depend on $\mathcal{A}$, but not on its realized random choices.

Proof of Lemma 11. We use the random-sign argument of Hazan, Agarwal, and Kale [29, Sec. 2.3, footnote 3]; see also [27, Sec. 3.2, Thm. 3.2]. Kumar, Dean, and Kleinberg [8, App. D, Thm. D.2] use a related construction with signs assigned to blocks for discounted memory. Here, each time has an independent sign, and the SDAC policies are evaluated on their own feasible storage trajectories.

Fix a causal feasible controller $\mathcal{A}$. Let $\xi_1, \dots, \xi_T$ be independent Rademacher random variables, independent of the controller's randomization. Set

$$w_t = 1, \qquad c_t(x,u) = \xi_t(L_x x - L_u u), \qquad t \in [T]. \tag{139}$$

Each cost is affine with gradient $\xi_t(L_x, -L_u)$, so it is convex and satisfies (9) because $|\xi_t| = 1$. The triangle inequality verifies (7):

$$\begin{aligned} |c_t(x,u) - c_t(x',u')| &\overset{(139)}{=} |L_x(x-x') - L_u(u-u')| \\ &\le L_x|x-x'| + L_u|u-u'|. \end{aligned}$$

Consider the two fixed SDAC parameters

$$\mathbf{m}^{(0)} = \mathbf{0}, \qquad \mathbf{m}^{(1)} = \mathbf{e}_1 = (1, 0, \dots, 0) \in \mathcal{M}_H. \tag{140}$$

For $t = 2, \dots, T$, (15), (1), (3), and (139) give

$$\begin{aligned} &u_t(\mathbf{m}^{(0)}) = 0, \qquad u_t(\mathbf{m}^{(1)}) = \alpha, \\ &x_t(\mathbf{m}^{(0)}) = \sum_{s=0}^{t-2} \alpha^s = \frac{1-\alpha^{t-1}}{1-\alpha}, \\ &x_t(\mathbf{m}^{(1)}) = 1. \end{aligned} \tag{141}$$

The state formula for $\mathbf{m}^{(1)}$ in (141) follows by induction from $x_2(\mathbf{m}^{(1)}) = 1$ and $\alpha \cdot 1 - \alpha + 1 = 1$. Subtracting the states in (141) gives

$$\begin{aligned} x_t(\mathbf{m}^{(0)}) - x_t(\mathbf{m}^{(1)}) &= \frac{1-\alpha^{t-1}}{1-\alpha} - 1 \\ &= \frac{\alpha(1-\alpha^{t-2})}{1-\alpha}. \end{aligned} \tag{142}$$

By (1), (15), and (139), both policies incur zero cost at $t = 1$. Thus, (11), (139), and the actions in (141) give

$$\begin{aligned} &J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)}) \\ &\quad= \sum_{t=2}^{T} \xi_t \left[L_x\big(x_t(\mathbf{m}^{(0)}) - x_t(\mathbf{m}^{(1)})\big) + \alpha L_u \right] \\ &\quad\overset{(142)}{=} \sum_{t=2}^{T} \xi_t \left(\frac{\alpha L_x(1-\alpha^{t-2})}{1-\alpha} + \alpha L_u \right) \\ &\quad\overset{(138)}{=} \sum_{t=2}^{T} \xi_t \beta_t. \end{aligned} \tag{143}$$

By causality, (x_t, u_t) depends only on past signs and the controller's randomization, so it is independent of ξ_t. By (10), (139), and $\mathbb{E}[\xi_t] = 0$,

$$\begin{aligned} &\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[J_T^{\mathrm{on}}] \\ &\quad= \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\big[\xi_t(L_x x_t - L_u u_t)\big] \\ &\quad= \sum_{t=1}^{T} \mathbb{E}[\xi_t]\, \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[L_x x_t - L_u u_t] = 0. \end{aligned} \tag{144}$$

The trajectories in (141) are deterministic. Hence, for $j \in \{0,1\}$, (11) and (139) give

$$\begin{aligned} &\mathbb{E}_{\boldsymbol{\xi}}[J_T(\mathbf{m}^{(j)})] \\ &\quad= \sum_{t=1}^{T} \mathbb{E}[\xi_t]\big(L_x x_t(\mathbf{m}^{(j)}) - L_u u_t(\mathbf{m}^{(j)})\big) = 0. \end{aligned} \tag{145}$$

Since both parameters in (140) belong to $\mathcal{M}_H$, (12) gives

$$\begin{aligned} \mathrm{Reg}_T^{\mathrm{SDAC}}(H) &\ge J_T^{\mathrm{on}} - \min\{J_T(\mathbf{m}^{(0)}), J_T(\mathbf{m}^{(1)})\} \\ &= J_T^{\mathrm{on}} - \frac{J_T(\mathbf{m}^{(0)}) + J_T(\mathbf{m}^{(1)})}{2} \\ &\quad + \frac{1}{2}\left| J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)}) \right|. \end{aligned} \tag{146}$$

The $p = 1$ Khintchine inequality [28] applies to the deterministic coefficients $\beta_2, \dots, \beta_T$ and independent signs:

$$\mathbb{E}_{\boldsymbol{\xi}} \left| \sum_{t=2}^{T} \xi_t \beta_t \right| \ge \frac{1}{\sqrt{2}} \sqrt{\sum_{t=2}^{T} \beta_t^2}. \tag{147}$$

Taking expectations in (146) and using (144) and (145) gives

$$\begin{aligned}\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[\mathrm{Reg}_T^{\mathrm{SDAC}}(H)] &\geq \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left|J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)})\right| \\ &\overset{(143)}{=} \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left|\sum_{t=2}^{T}\xi_t\beta_t\right| \\ &\overset{(147)}{\geq} \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2}.\end{aligned} \tag{148}$$

The expectation in (148) averages over the 2^T sign sequences. Therefore, some $\boldsymbol{\xi}^\star$ satisfies

$$\begin{aligned}\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}(H) \mid \boldsymbol{\xi} = \boldsymbol{\xi}^\star\big] &\geq \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[\mathrm{Reg}_T^{\mathrm{SDAC}}(H)] \\ &\overset{(148)}{\geq} \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2}.\end{aligned} \tag{149}$$

Fix $\boldsymbol{\xi}^\star$ in (139) before the interaction. This gives a deterministic oblivious sequence, and (149) proves (137). □

Proof of Theorem 2. Since $0 < \alpha < 1$ and $\log\alpha = -\log(1/\alpha)$, for $t = \lceil \log 2/\log(1/\alpha)\rceil + 2, \dots, T$,

$$\begin{aligned}\alpha^{t-2} &\leq \alpha^{\lceil \log 2/\log(1/\alpha)\rceil} \\ &\leq \alpha^{\log 2/\log(1/\alpha)} = e^{-\log 2} = \frac{1}{2}.\end{aligned} \tag{150}$$

By (138) and $L_x, L_u > 0$,

$$\begin{aligned}\beta_t &\overset{(150)}{\geq} \alpha\left(L_u + \frac{L_x}{2(1-\alpha)}\right) \\ &\geq \frac{\alpha}{2}\left(L_u + \frac{L_x}{1-\alpha}\right) \overset{(23)}{=} \frac{G_\alpha}{2}.\end{aligned} \tag{151}$$

The number of indices in (150) satisfies

$$T - \left\lceil \frac{\log 2}{\log(1/\alpha)}\right\rceil - 1 \overset{(51)}{\geq} \frac{T}{2}. \tag{152}$$

Discarding nonnegative squares and applying (151),

$$\begin{aligned}\sum_{t=2}^{T}\beta_t^2 &\geq \sum_{t=\lceil \log 2/\log(1/\alpha)\rceil+2}^{T}\beta_t^2 \\ &\geq \left(T - \left\lceil \frac{\log 2}{\log(1/\alpha)}\right\rceil - 1\right)\frac{G_\alpha^2}{4} \\ &\overset{(152)}{\geq} \frac{TG_\alpha^2}{8}.\end{aligned} \tag{153}$$

For the sequence supplied by Lemma 11, $G_\alpha > 0$ and (153) yield (52):

$$\begin{aligned}\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}(H)\big] &\overset{(137)}{\geq} \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2} \\ &\overset{(153)}{\geq} \frac{1}{2\sqrt{2}}\sqrt{\frac{TG_\alpha^2}{8}} \\ &= \frac{G_\alpha}{8}\sqrt{T}.\end{aligned}$$

□

Justification of Remark 2. For fixed $L_x, L_u > 0$, (23) gives

$$(1-\alpha)G_\alpha = \alpha\big(L_x + (1-\alpha)L_u\big) \longrightarrow L_x \tag{154}$$

as $\alpha \uparrow 1$. Since $1 \leq z^{-1} \leq \alpha^{-1}$ for $z \in [\alpha, 1]$,

$$\begin{aligned}1 \leq \frac{\log(1/\alpha)}{1-\alpha} &= \frac{1}{1-\alpha}\int_\alpha^1 \frac{dz}{z} \\ &\leq \frac{1}{\alpha} \longrightarrow 1.\end{aligned} \tag{155}$$

By (154) and (155), both G_α and the right-hand side of (51) are of order $(1-\alpha)^{-1}$; taking a ceiling changes a number by less than one. Under (51), (52) and (154) imply

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}(H)\big] = \Omega\left(\frac{\sqrt{T}}{1-\alpha}\right). \tag{156}$$

For the upper bound, (43) and (154) give

$$\begin{aligned}(1-\alpha)^3\Gamma_\alpha &= \frac{1-\alpha}{2}\big((1-\alpha)G_\alpha\big)^2 \\ &\quad + \alpha\big((1-\alpha)G_\alpha\big)(L_x + \alpha L_u) \\ &\longrightarrow L_x(L_x + L_u) > 0.\end{aligned} \tag{157}$$

By (157),

$$\sqrt{\Gamma_\alpha} = \Theta\Big((1-\alpha)^{-3/2}\Big) \quad \text{as } \alpha \uparrow 1. \tag{158}$$

Together with (45), this gives the upper bound stated in Remark 2.

For fixed T and $t = 2, \dots, T$,

$$\frac{1-\alpha^{t-2}}{1-\alpha} = \sum_{s=0}^{t-3}\alpha^s \longrightarrow t-2 \qquad \text{as } \alpha \uparrow 1, \tag{159}$$

where the sum is empty for $t = 2$. Equations (159) and (138) give $\beta_t \to L_u + L_x(t-2)$ for every $t = 2, \dots, T$. Since this is a finite set of indices, the right-hand side of (137) remains finite. □

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